

Project 1: Student Result Management System

Objective

Develop a C program to manage the result of students by calculating total marks, percentage, grade, and pass/fail status.

Problem Statement

Create a menu-driven program that allows the user to enter a student's details and display the result.

Requirements

Your program should perform the following operations:

1. Enter Student Name
2. Enter Roll Number
3. Enter Marks of 5 Subjects
4. Calculate Total Marks
5. Calculate Percentage
6. Assign Grade using:
 - A → 90% and above
 - B → 80% – 89%
 - C → 70% – 79%
 - D → 60% – 69%
 - F → Below 60%
7. Display Pass or Fail (Pass marks = 40 in every subject)

Sample Output

***** STUDENT RESULT *****

Enter Student Name : Rahul

Enter Roll Number : 101

Enter Marks of English : 85

Enter Marks of Mathematics : 92

Enter Marks of Science : 78

Enter Marks of Computer : 95

Enter Marks of Hindi : 80

Student Name : Rahul

Roll Number : 101

Total Marks : 430

Percentage : 86.00%

Grade : B

Result : PASS

Concepts Used

- Variables
- if-else
- Arrays
- Functions
- Loops

Project 2: ATM Banking System

Objective

Develop a simple ATM application that performs basic banking operations.

Problem Statement

Write a menu-driven C program that allows the user to perform ATM transactions.

Requirements

Initial Balance = ₹10,000

Menu:

1. Check Balance
2. Deposit Money
3. Withdraw Money

4. Exit

Rules:

- Withdrawal amount cannot exceed balance.
- Deposit amount must be positive.
- Display updated balance after every transaction.

Sample Output

===== ATM MENU =====

1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit

Enter Choice : 2

Enter Deposit Amount : 2500

Amount Deposited Successfully.

Current Balance : ₹12500

Another Example

Enter Choice : 3

Enter Withdrawal Amount : 15000

Insufficient Balance!

Current Balance : ₹12500

Concepts Used

- switch-case
 - while loop
 - Functions
 - Variables
-

Project 3: Grocery Billing System

Objective

Create a billing system that calculates the total amount of purchased grocery items.

Problem Statement

Write a C program that generates a grocery bill based on the quantity and price of purchased items.

Requirements

- Enter Customer Name
- Enter Number of Items
- For each item:
 - Item Name
 - Quantity
 - Price per Unit
- Calculate Total Price of each item
- Calculate Grand Total
- Display formatted bill

Sample Output

===== GROCERY BILL =====

Customer Name : Aman

Enter Number of Items : 3

Item 1

Name : Rice

Quantity : 2

Price : 60

Item 2

Name : Sugar

Quantity : 3

Price : 45

Item 3

Name : Milk

Quantity : 5

Price : 30

Item	Qty	Price	Total
Rice	2	60	120
Sugar	3	45	135
Milk	5	30	150
Grand Total		₹405	

Thank You!

Concepts Used

- Arrays
- Strings
- Loops
- Functions

Submission Guidelines

Students should:

1. Write the program in C.
2. Use meaningful variable names.
3. Divide the program into functions wherever possible.
4. Display proper menus and messages.
5. Test the program with different inputs.
6. Add comments to explain important parts of the code.